

Event Manager Project Documentation

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Project: Event Manager

1. Project Overview

Event Manager is a full-stack web application built with Next.js, TypeScript, Prisma, and SQLite. The application allows users to register, log in, create events, view events, edit and delete their own events, and open detailed event pages.

The main resource of the application is an event. Each event contains a title, description, location, date, creation date, update date, and author information.

2. Technologies Used

- Next.js App Router for routing and server actions.
- TypeScript for typed application code.
- React for the user interface.
- Prisma ORM for database models, migrations, and queries.
- SQLite for the local development database.
- Tailwind CSS for styling and responsive layout.
- bcryptjs for password hashing.
- Cookie-based authentication for user sessions.

3. Main Features

- User registration with username, email, and password.
- Login and logout functionality.
- Password hashing before storing passwords in the database.
- Guests can view public events and event details.
- Logged-in users can create new events.
- Users can edit and delete only their own events.
- Events include title, description, location, date, createdAt, and updatedAt fields.
- Validation messages are displayed for invalid or missing input.
- Dynamic event details pages.
- Protected routes for authenticated users.

4. Database Structure

The database is managed with Prisma ORM.

User model:

- username
- email
- hashed password
- createdAt
- events created by the user

Event model:

- title
- description
- location
- date
- createdAt
- updatedAt
- authorId
- relation to the user who created the event

5. Application Pages

- Home Page (/)
- Register Page (/register)
- Login Page (/login)
- Events Page (/events)
- Create Event Page (/events/new)
- Event Details Page (/events/[id])
- Edit Event Page (/events/[id]/edit)

6. How to Run the Project

1. Open the project folder in a terminal.
2. Install dependencies with: `npm install`
3. Create a `.env` file in the project root.
4. Add:
`DATABASE_URL="file:./dev.db"`
`JWT_SECRET="WHERE'S_OMNI_MAN_WHERES_HE?"`
5. Run Prisma migrations:
`npx prisma migrate dev --name init`
6. Generate Prisma Client:
`npx prisma generate`
7. Start the development server:
`npm run dev`
8. Open `http://localhost:3000`

7. Authentication Flow

When a user registers, the application validates the input, checks whether the email already exists, hashes the password using bcryptjs, stores the user in the database, and creates a session cookie.

Protected pages and actions check the cookie to identify the current user.

8. CRUD Flow

- Create: users can create events.
- Read: all users can view events.
- Update: only event owners can edit events.
- Delete: only event owners can delete events.

9. Validation and Error Handling

- Empty required fields
- Invalid email addresses
- Short passwords
- Duplicate accounts
- Wrong login credentials
- Unauthorized access
- Missing records
- Invalid form submissions

10. Development Log

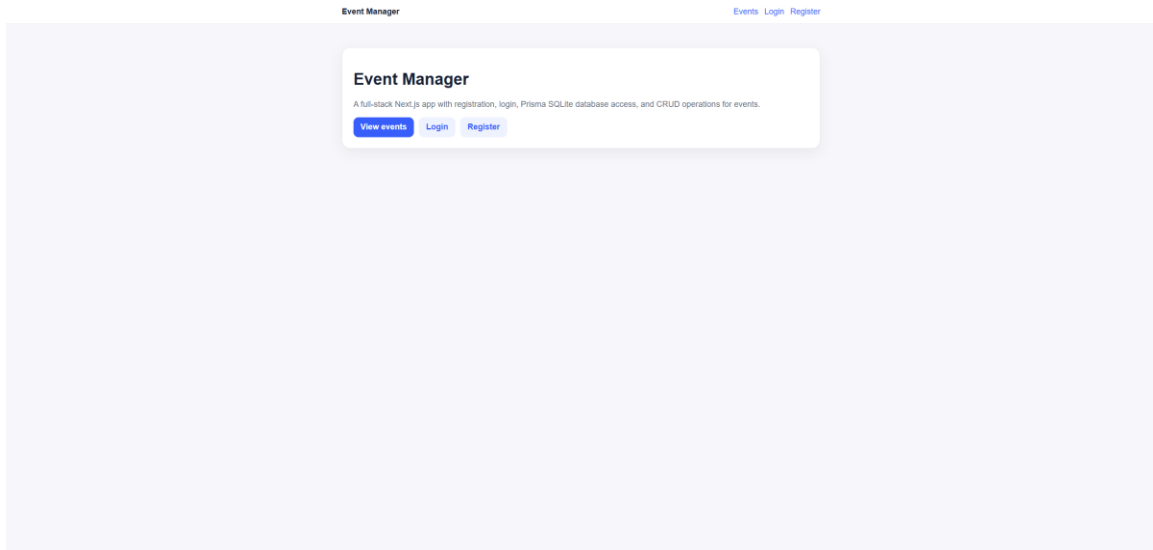
- Created the Next.js project structure.
- Configured Prisma and SQLite database.
- Added authentication system.
- Added Event CRUD functionality.
- Added validation and protected routes.
- Improved navigation and UI.
- Prepared project documentation.

11. Problems Solved

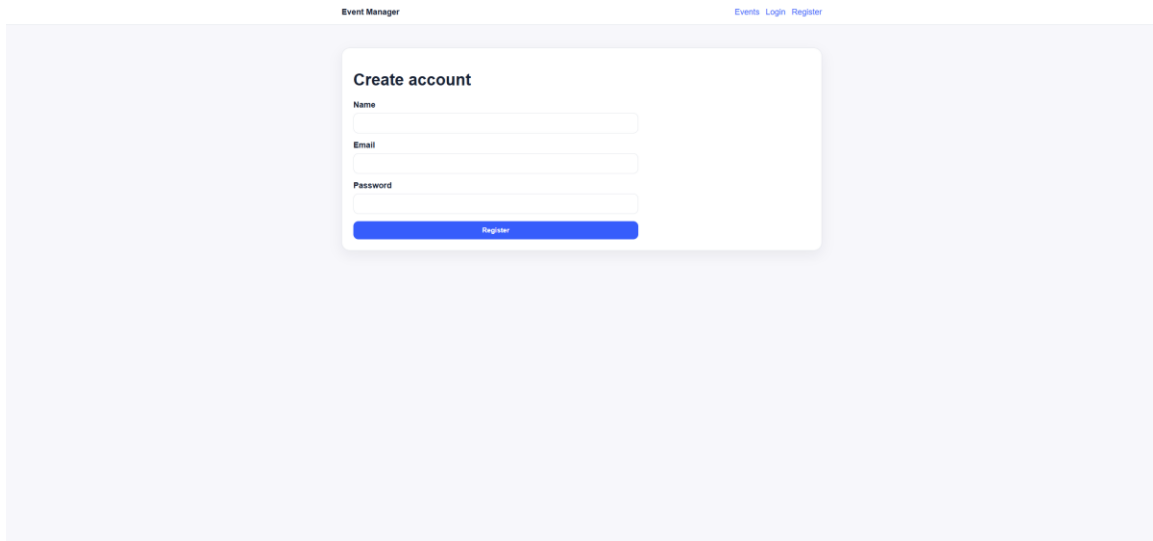
- Protected edit/delete actions.
- Added password hashing.
- Fixed Prisma environment variable issues.
- Added validation for forms.
- Protected private pages from unauthorized users.

12. Screenshots

Home Page



Register Page



Login Page

Event Manager

EventsLoginRegister

Login

Email

Password

Login

Events List Page

Event Manager

EventsCreate EventHi, TESTLogout

EventsCreate event

WHERE'S OMNI MAN WHERE'S HE?

11/13/2006 - IMMORTAL

OMNI MAAAAAAAAAAAAAN

Created by TEST

View details

Create Event Page

Event Manager

EventsCreate EventH, TESTLogout

Create event

Title

Description

Date and time

mm/dd/yyyy --:-- --

Venue

Ticket price

Save event

Event Details Page

Event Manager

EventsCreate EventH, TESTLogout

WHERE'S OMNI MAN WHERE'S HE?

11/13/2006, 10:00:00 AM - IMMORTAL

OMNI MAAAAAAAAAAAAAN

Ticket price: \$5.00

Created by TEST

Back to list

Edit

Delete

Edit Event Page

The screenshot shows a web application interface for editing an event. At the top, there is a navigation bar with the text 'Event Manager' on the left and a series of links on the right: 'Events', 'Create Event', 'Hi, TEST', and 'Logout'. The 'Logout' link is highlighted with a blue background. Below the navigation bar, the main content area is a light gray. In the center, there is a white card titled 'Edit event'. Inside the card, there are several form fields: 'Title' with the value 'WHERE'S OMNI MAN WHERE'S HE?', 'Description' with the value 'OMNI MAAAAAAAAAAAAAN', 'Date and time' with the value '11/13/2006 10:00 AM' and a calendar icon, 'Venue' with the value 'IMMORTAL', and 'Ticket price' with the value '5'. At the bottom of the card, there is a blue button labeled 'Update event'.

13. Conclusion

The project successfully meets the requirements for a full-stack Next.js application with authentication, Prisma integration, CRUD functionality, dynamic details pages, validation, and protected routes.